Aarav Shah

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ABOUT ME

I build intelligent systems that solve real problems—from enabling computer access for individuals without hands to designing quantum-inspired supply chain optimization. My work sits at the intersection of applied machine learning, operations research, and human-centered technology, driven by an obsession with turning theoretical breakthroughs into tools that create measurable impact.

ACADEMIC PROFILE

University of California, Riverside
Research Focus: Applied Machine Learning, Operations Research, Supply Chain Intelligence

LEADERSHIP & SIGNIFICANT ACTIVITIES

Gesture-Controlled Computing System | Developer

2022 • 40 hrs/week • 40 total hours
Aligns with MIT's: Innovation, Hands-on Learning

Faced with the challenge of creating accessible computing interfaces during a 24-hour hackathon at Indus International School, I engineered a system enabling cursor control through hand and finger gestures detected via webcam. Built the computer vision pipeline from scratch, implementing real-time gesture recognition algorithms that translated movement into precise cursor commands. Secured 1st place among 50+ competing teams, with judges citing the project's potential for accessibility applications.

Key Outcomes:
- 1st place finish against 50+ teams
- Real-time gesture recognition with <100ms latency
- Foundation for later accessibility-focused development

AWS Data Science Program | Team Leader

December 2023 • 40 hrs/week • 80 total hours
Aligns with MIT's: Collaboration, Problem Solving

Selected for National University of Singapore's intensive data science cohort, I led a 4-member team tackling customer churn prediction. Mastered Random Forest, SVM, KNN, and Neural Network architectures, then architected an ensemble approach combining multiple models. Achieved 99.8% prediction accuracy—the highest in our cohort—while mentoring teammates through AWS infrastructure deployment.

Key Outcomes:
- 99.8% model accuracy (top performer in cohort)

• Led team of 4 to successful project completion
• First in cohort to deploy production models on AWS

DSA Engineering Internship | Software Engineering Intern

June–August 2024 • 40 hrs/week • 320 total hours
Aligns with MIT's: Problem Solving, Hands-on Learning

Assigned to optimize legacy codebases at a software firm, I refactored 15+ modules using advanced data structures and algorithms. Reduced average query response time by 40% through implementing efficient tree traversal and graph algorithms. Solved 200+ algorithmic challenges, translating theoretical CS concepts into production-ready solutions.

Key Outcomes:
• 40% reduction in query response times
• 15 modules refactored and optimized
• 200+ algorithmic problems solved

IBM Full Stack Development Certification | Developer

160 total hours
Aligns with MIT's: Hands-on Learning, Innovation

Completed IBM's rigorous certification while building 5 production-grade web applications. Mastered React, Node.js, Express, MongoDB, and CI/CD pipelines, deploying applications to IBM Cloud. Created a portfolio management system handling 10,000+ data points with sub-second response times.

Key Outcomes:
• 5 full-stack applications deployed to production
• Proficiency across 12+ technologies
• CI/CD pipeline implementation for automated deployment

Accessibility Computing Application | Lead Developer

200 total hours
Aligns with MIT's: Innovation, Impact

Recognizing that millions of individuals with upper limb disabilities lack computer access, I engineered an application combining nose-movement tracking with speech recognition for complete hands-free computer control. Built custom ML models for facial landmark detection achieving 95% accuracy, integrated speech-to-command functionality for mouse clicks. Currently preparing for open-source release to maximize accessibility impact.

Key Outcomes:
• 95% accuracy in nose-movement cursor control

• Speech recognition integration for click commands
• Designed for open-source distribution

Daricha Organization | Curriculum Manager

April 2024–Present • 25 hrs/week • 1,950 total hours
Aligns with MIT's: Impact, Collaboration

Leading educational program design for 200+ Afghan girls denied schooling under Taliban rule, I created comprehensive STEM and literacy curricula adapted for remote, resource-constrained environments. Recruited and trained 12 volunteer instructors, established assessment frameworks tracking student progress, and increased program enrollment by 150% through community outreach.

Key Outcomes:
• 200+ students served across Afghanistan
• 12 instructors recruited and trained
• 150% enrollment growth in 18 months

Rotary Club Community Service | Active Member

September 2022–September 2024 • 4 hrs/week • 208 total hours
Aligns with MIT's: Impact, Collaboration

Organized 8 donation drives collecting 2,000+ items for underserved families, coordinated logistics for community outreach events reaching 500+ residents. Designed awareness campaigns on education access that secured $3,000 in local donations for scholarship funds.

Key Outcomes:
• 2,000+ items collected and distributed
• 500+ community members reached
• $3,000 raised for scholarships

ARIMA Inventory Forecasting Research | Researcher

IEEE • 20 hrs/week • 520 total hours
Aligns with MIT's: Intellectual Curiosity, Innovation

Investigated ARIMA model applications for inventory optimization, analyzing 3 years of historical sales data across 500+ SKUs. Developed forecasting models reducing prediction error by 23% compared to baseline methods. Research accepted for IEEE publication, establishing foundation for subsequent quantum-enhanced supply chain work.

Key Outcomes:
• 23% improvement in forecast accuracy
• 500+ SKUs analyzed across 3-year dataset
• IEEE publication acceptance

Quantum-Enhanced Supply Chain Research | Lead Researcher

2,080 total hours

Aligns with MIT's: Innovation, Intellectual Curiosity

Authored comprehensive research integrating AI forecasting, IoT sensing, blockchain verification, and SAP S4HANA ERP into a unified supply chain architecture. Proposed self-regulating systems achieving 18% cost reduction in simulations while improving delivery reliability by 25%. Framework now under review for enterprise pilot implementation.

Key Outcomes:

- 18% simulated cost reduction
- 25% improvement in delivery reliability metrics
- Enterprise pilot under consideration

Quantum Decision Theory for Retail | Researcher

Ongoing

Aligns with MIT's: Innovation, Problem Solving

Pioneered unified framework combining ARIMA forecasting, quantum decision theory, and reinforcement learning for omnichannel retail optimization. Modeled customer intent as quantum superpositions capturing cross-channel dependencies, achieving 12–18% profitability improvement and 30% stock-out reduction in simulations versus classical approaches.

Key Outcomes:

- 12–18% profitability improvement demonstrated
- 30% reduction in stock-outs
- Novel quantum-classical hybrid methodology

F1 Race Insights | Founder & Lead Developer

Applicable AI • Ongoing

Aligns with MIT's: Innovation, Hands-on Learning

Built end-to-end ML platform predicting Formula 1 race outcomes with 98.7% AUC-ROC accuracy—14x better than random chance. Engineered 68 features from 1,000+ historical races, implemented 8 ML models including custom neural architecture (NBT-TLF), and deployed 10,000-iteration Monte Carlo simulations for championship projections. Full-stack system includes Next.js frontend, FastAPI backend, Docker containerization, and AWS deployment with MLOps monitoring.

Key Outcomes:

- 98.7% AUC-ROC prediction accuracy
- 8 ML models benchmarked and deployed
- 10,000 Monte Carlo simulations per projection

SAP AI Implementation | Applied AI Engineer

May–December 2025 • 20 hrs/week • 500 total hours
Aligns with MIT's: Impact, Problem Solving

Deployed ML models into SAP's enterprise workforce applications serving millions of users. Translated experimental models into production services with monitoring, logging, and rollback mechanisms. Reduced model drift incidents by 35% through implementing real-time performance evaluation, iterating based on live operational feedback rather than offline metrics.

Key Outcomes:
• Models deployed to millions of enterprise users
• 35% reduction in model drift incidents
• Production ML monitoring framework established

Hyperspectral Imaging Research | Research Assistant

UC Riverside • Ongoing
Aligns with MIT's: Intellectual Curiosity, Innovation

Conducting comparative analysis of tensor decomposition methods (SVD, CP, Tucker) versus neural implicit representations for hyperspectral data compression. Designed controlled experiments across compression ratios and noise levels, identifying regimes where classical methods outperform neural approaches in stability. Established reproducible evaluation framework for remote sensing applications.

Key Outcomes:
• Comparative analysis across 5+ compression methods
• Reproducible evaluation framework created
• Guidance for method selection in imaging tasks

Inventory Quantum Platform | Founder & Lead Developer

January 2026–Present
Aligns with MIT's: Innovation, Hands-on Learning

Architected enterprise-grade supply chain platform integrating Google's Temporal Fusion Transformer (94.8% forecast coverage), quantum-inspired QUBO optimization, and offline reinforcement learning. Built S3-backed cloud infrastructure, real-time drift detection, and LLM-powered analytics copilot. System achieves sub-5% demand forecast error rates with 95% coverage guarantees through conformal prediction.

Key Outcomes:
• 94.8% forecast coverage with conformal guarantees
• Sub-5% demand prediction error rates
• Full quantum-classical optimization pipeline

Student Council | Vice President

September 2019–June 2020 • 5 hrs/week • 200 total hours
Aligns with MIT's: Collaboration, Impact

Managed $15,000 annual budget for school events serving 1,200 students. Organized 12 major events including fundraisers generating $4,000 for local charities. Represented student body in 20+ administrative meetings, successfully advocating for extended library hours and new computer lab equipment.

Key Outcomes:
• $15,000 budget managed
• $4,000 raised for charity
• Policy changes benefiting 1,200 students

School Reforestation Initiative | Project Lead

2 total hours
Aligns with MIT's: Impact

Coordinated tree-planting effort on school campus, organizing 25 student volunteers to plant 40 trees in a single day, increasing campus green coverage by 15%.

Green Tokri Sustainable Agriculture | Participant

40 total hours
Aligns with MIT's: Impact, Hands-on Learning

Completed 20-week organic farming program, implementing water-efficient irrigation techniques that reduced resource usage by 30% in school garden. Applied sustainable practices to maintain 40+ plants throughout academic year.

Youth Mentorship Program | Mentor

80 total hours
Aligns with MIT's: Collaboration, Impact

Mentored 8 middle school students in mathematics and computer science fundamentals over 40 weeks. 6 of 8 mentees improved grades by one letter grade or more; 3 subsequently enrolled in advanced STEM courses.

Disaster Relief Volunteering | Volunteer

8 total hours
Aligns with MIT's: Impact

Supported emergency response logistics during regional flooding, helping coordinate supply distribution to 50+ affected families.

Environmental Cleanup Campaign | Volunteer

8 total hours
Aligns with MIT's: Impact

Organized campus cleanup removing 200+ pounds of waste, recruiting 15 volunteers for sustainability initiative.

HONORS & RECOGNITION

• **Silver Medal — 5th International STEM Olympiad** (2025)
Global recognition among top performers in science, technology, engineering, and mathematics

• **National-Level Chess Player** — Top 200 in India (Under-19)
2-time state representative, 2-time district champion; competing since age 8

• **IEEE Research Publication** — Quantum-Enhanced Retail Operations Framework (2025)
Novel integration of ARIMA forecasting with quantum decision theory

• **Research Recognition** — Mathematical Foundations of AI in Scientific Discovery (2025)
Unified framework bridging drug discovery, materials science, and fundamental physics

TECHNICAL SKILLS

Machine Learning: TensorFlow, PyTorch, XGBoost, CatBoost, LightGBM, Temporal Fusion Transformers, Reinforcement Learning (CQL), SHAP Explainability
Languages: Python, JavaScript/TypeScript, SQL, R
Frameworks: React, Next.js, Node.js, FastAPI, Express.js
Infrastructure: AWS, Docker, CI/CD, MLOps, S3, IBM Cloud
Research: Tensor Decomposition, Quantum Optimization (QUBO/QAOA), Time Series Forecasting, Computer Vision
Languages: English (Native), Hindi (Native)

WHY MIT

MIT's integration of rigorous theory with hands-on building mirrors my approach—every research project I undertake results in deployed, functional systems. I'm drawn to the Operations Research Center's work on supply chain optimization, where my quantum-enhanced forecasting research could contribute to Professor Bertsimas's prescriptive analytics initiatives. The MIT-IBM Watson AI Lab's focus on enterprise AI deployment aligns directly with my SAP implementation experience. Most importantly, MIT's culture of "mens et manus" reflects my conviction that the best ideas prove themselves through working systems that create measurable impact.

ADDITIONAL EXPERIENCE

Researcher | IEEE

Unknown - Unknown • 520 total hours

Conducted research on using ARIMA (Autoregressive Integrated Moving Average) models for inventory forecasting. Analyzed historical data, developed forecasting models, and evaluated the accuracy and effectiveness of the ARIMA approach.

Researcher |

- Present • 0 total hours

Proposes a unified framework combining ARIMA demand forecasting, quantum decision theory, and reinforcement learning for omnichannel retail.

Models customer intent as quantum superpositions to capture contextual and cross-channel dependencies absent in classical models.

Embeds forecast uncertainty directly into decision states and reward structures.

Demonstrates improved profitability (12–18%), reduced stock-outs, and higher service levels under demand volatility and promotions.

Researcher |

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